

Table of Contents

<u>1. Android studio setup & Guide</u>	<u>Page - 1</u>
<u>2. Recyclerview</u>	<u>Page - 2</u>
<u>3. Recyclerview model & QR scanner</u>	<u>Page-3</u>
<u>4. Server setup apache tomcat, mysql, Postman API any changes/problems & Volley library dependency</u>	<u>Page 4-7</u>
<u>5. Servlet and Volley library</u>	<u>Page 8-9</u>

Installation of Android Studio:-

Initially installed the latest version of android studio but was not able to sync gradle else everything was possible later tried with older versions where it was asking about proxy settings so tried installing/ syncing gradle with wifi/ without proxy network then it worked and working efficiently.

To download android studio -> simply type android studio installation on google and it will let us install latest stable android studio latest asking for all permissions and download sdk stuffs letting it allow after checking the available space in system. Also downloading the latest version of Java (JDK).

Creating Empty views activity:

When we install android studio, it takes quite time and after that it will ask for project so taking the empty views activity and the suitable android version which can be used in any device and moving with empty views activity will let us make the UI as well as code part involved JAVA/Kotlin. Gradle sync will take place and we are ready further to make our customized project. Also choosing relative layout/setting it in XML files, as constraint layout creates problems.

Creating an emulator(Android studio's device to see application running) :

We can setup this by moving to right top side and device there on clicking + sign it will show us remote/virtual device and selecting virtual device. Further choosing our device of choice I choosed google pixel 8a. Later whatever things it will install related to emulator and our device will be ready on initial run of program it will start device.

To pair with your mobile device :

Enabling the developer setting, which will be enabled in settings>about phone & later about model tap for 5 times, you can see the video on youtube for this if not proceeded with this. Next step, clicking wireless debugging and clicking it on, This setup will happen when both devices are connected to same wifi or device's hotspot is used for PC.

On Android studio going to device option right top side and the one with + and wifi signal around it choosing it, further it will ask for QR or pair with code, same we will get and device and later it can be paired.

Below we can see about setup of recyclerview due to which our list is available.

Why & How to proceed with RecyclerView :

The representation of list with listview is now old version so using recyclerview is the new version to represent it.

For recyclerview we cannot use the same empty view activity having xml and java file, in XML file we will just include the headings and use recyclerview thing there but to bind the list in proper manner we must create a resource file in layout and naming it as -> name_row and it will be xml file, A simple plain file with our column sizes we made and their IDs no background or no heading for that columns.

How to create adapter for RecyclerView:-

Detail can be found from 46.28 of this video link, Below steps are written though from start https://youtu.be/FEqF1_jDV-A?si=pPkFWylchBwAEGW7

- Firstly making a java class with name, here mostly **RecyclerviewwhatevernameAdapter.**
- After java file is created, later writing **extends** after the
public class RecyclerviewwhatevernameAdapter
- After extends
RecyclerView.Adapter<RecyclerviewwhatevernameAdapter.ViewHolder>
- This will show error on ViewHolder or complete class, right clicking on ViewHolder and selecting create class ViewHolder -> This will create class for ViewHolder yet show error on old one class RecyclerviewwhatevernameAdapter
- Further, clicking on the right option on original class of file and clicking on create ViewHolder extends widget from available options, it will extend the class ViewHolder.
- Later it will show error on class ViewHolder and then right clicking on (ViewHolder class) it and clicking on create constructor matching super, which will create constructor for ViewHolder.
- Then again getting back to original class of generated file (which was 1st class in file), again right click and more context and then clicking on implement methods to further implement available all 3 methods.
- Further whatever work we need to do is available into this 3 methods created, we can customize according to our need.

Creating model file for list items :-

To include all the items in the list we need to make that particular list model and make the mapping like this.id = id... and getters into that model file. Just like this

```
public item_list_model(int id, String code, String name) {
    this.id = id;
    this.code = code;
    this.name = name;
}

public int getId() {
    return id;
}
```

Creating QR scanner in project :-

- With integrating this dependencies in build.gradle.kts(app) -

```
implementation("com.google.zxing:core:3.4.1")
implementation("com.journeyapps:zxing-android-embedded:4.2.0")
```

- Later inserting this permissions inside manifests.xml at top of available project details

```
<uses-permission android:name="android.permission.CAMERA"/>
<uses-feature android:name="android.hardware.camera"/>
```

- later going on with xml part and java file

My QR was in landscape to move it to portrait

<https://youtu.be/mdpxaLwzPHg?si=PK-83mu-JU3R1AtV>

```
<activity android:name="com.journeyapps.barcodescanner.CaptureActivity"
    android:screenOrientation="portrait"
    tools:replace="screenOrientation"/>
```

Add above In manifest.xml file

Setting up web server with ApacheTomcat, MySQL & Eclipse IDE :

Things needed to setup :-

https://youtu.be/yyIfROdULw4?si=crJ7lf3zZqN_zqHk

MySQL application/ MySQL with command prompt - downloading it from chrome.

Eclipse IDE with Developer one(**Apache Tomcat 10** server) - downloading 1st Eclipse IDE and setting it up with tomcat server 10 which will get downloaded into local folder at selected location. Using latest stable JDK 23/21.

PostMan API - Just download for windows version from chrome to test sevlet/API.

MySQL-Java connector jar file :- Inserting the connector jar file into apache tomcat server > library folder.

Android studio in use already.

Android studio Internet permission

- Syncing the gradle and then in manifests section -> AndroidManifest.xml file in files top side, we can add the permission to use internet after the top declaration
<Manifest..../>
- <uses-permission android:name="android.permission.INTERNET" />

If any problem with Tomcat installation : -

https://youtu.be/9RiQoRdgGGE?si=L9C9_GqBKAatp5dA

. Proceeding with downloads when checklist comes make sure to check all option & then proceed further.

- . If already have used port 8080 better to use any other port like one used by me 8082 (or 9494 while setting apache tomcat.)
- . Note down username & password for apache tomcat.(admin & admin1234) & remember location of file or note down it. Finishing with installation.
- .running can be checked by running the port used by us here localhost://8082

New project in NetBeans IDE/ Eclipse IDE —

- New project -> java with ant -> java web -> web application
- proceeding with suggested location storage -> later moving on to selecting tomcat in next stage selecting the comfortable name of project here (Applicationdemo1)
- In source package -> making java package with name -> com.example.Applicationdemo1
- creating new servlet inside it by rightclick -> new -> servlet , setting name and getting with next and finish.(here DBAPI)
- In loaded file, removing all part inside try(...) function from protected void process request and keeping the part inside () of try function.

Given below things to implement further in here, importing the conenction class and driver manager properly -> `java.sql.Connection, java.sql.DriverManager`

protected void processRequest(HttpServletRequest request, HttpServletResponse response)

throws ServletException, IOException {

PrintWriter out = response.getWriter();

String username =request.getParameter("name");

String email = request.getParameter("email");

String password =request.getParameter("password") ;

try{

Class.forName("com.mysql.cj.jdbc.Driver");

Connection con =

DriverManager.getConnection("jdbc:mysql:<http://localhost:3306/dbname>","dbusername",["password"](http://localhost:3306/dbname));

```

    }catch(Exception e){
        out.print(e);
    }
}

```

Whenever using tomcat server/jakarta/ java SE:-

Making sure the perfect versions of above mentioned things match with each other, otherwise it can create a problem here I used tomcat server 10 & latest version Eclipse IDE & default JAVA SE is latest 23/21. Since jakarta is latest & java EE is old hence using jakarta is preferred.

- Making sure default server for tomcat server will be 8080 but it might cause problem apart from it we can choose any other whichever we like eg:-8082/8090.

- Whenever adding mysql-jdbc connector in eclipse make sure to add it in server lib by going to downloaded location of tomcat and then selecting lib folder and pasting inside it. Later refresh project and run, it will run this time. If ref lib of connector gets created then please remove it. Refer video - https://youtu.be/UzJb10SlA0?si=n_-D6K2tlCelzm

Connecting to same URL of servlet on mobile phone:-

- Making sure that our mobile and PC is connected to same network or mobile hotspot is connected to PC.
- Getting what is IP of it :- ipconfig on command prompt will let us know IP at IPv4 address
-
- going to wifi and then clicking on more about the connected one, then configure firewall & security settings and switching it off and then typing the IP and the further things after localhost: will get us to the desired API

When running application on mobile your data might show error like Cleartext HTTP traffic not permitted or error with volley library

Make sure to create new xml file in res/xml with name - network_security_config
Later in that xml file it must look like - here 192.168.0.100 is IP at that time

```
<?xml version="1.0" encoding="utf-8"?>
<network-security-config>
  <domain-config cleartextTrafficPermitted="true">
    <domain includeSubdomains="true">192.168.0.100</domain>
  </domain-config>
</network-security-config>
```

After this include this things in manifest file :- under section

```
<application android:networkSecurityConfig="@xml/network_security_config" ... >
</application>
```

Else :- Directly into manifest file in application section write
android:usesCleartextTraffic="true"

POSTMAN Api —

Pasting URL in new tab by signing in and it will show the expected response by our URL.

Opening MySQL on Command prompt:-

https://youtu.be/kj_oW8cx6Bs

- Open Command prompt -> trace the location for MySQL BIN and
- copy the path -> C:\Program Files\MySQL\MySQL Server 8.0\bin
- Onto command prompt -> cd "above copied path"
- Command prompt will change to ->
C:\Program Files\MySQL\MySQL Server 8.0\bin>
- Then type -> mysql -u *username* -p
- Further it will ask for password for the mysql server whatever set by us by entering it we will be directed to mysql.
- Later creating tables and using database as per SQL query.

Volley Library dependency :-

Adding this dependency in gradle.build.kts :-

implementation("com.android.volley:volley:1.2.1")

And then let the gradle sync.

Servlet format can be found on page 6 later half and similar format used for servlet coding .

8

More Information about servlet and Volley library:-

How to add new servlet in eclipse IDE, top left side> name of project(app1)>java resources> name of package(Package1)> right clicking and then new > servlet at lower options. This will create new servlet and removing unnecessary parts from servlet and using the above servlet format and using post, get requests.

Whenever starting servlet making sure the tomcat server is on/ restarted after doing necessary changes.

Whenever sending requests from Java file in android studio any `Request.Method.POST` or `Request.Method.GET` Can be used only of servlet includes this functions

@Override

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
```

@Override

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
```

Whenever we use URL in volley request making sure about this format :

http://IP-address: portNoOfTomcatServer/projectname/servletname
Like here for login page

http://192.168.0.103 :8082/app1/login

Connection con =

*DriverManager.getConnection("jdbc:mysql:<http://localhost:3306/dbname>","dbusername",
["password"](#)");*

Connection con =

**DriverManager.getConnection("jdbc:mysql://localhost:3306/signupinfo?useSSL=false&serverTime
zone=UTC&allowPublicKeyRetrieval=true","root","Viraj@9311");**

Servlet query

PreparedStatement ps = con.prepareStatement(

"UPDATE pick_order SET date = ?, name = ?, no_of_kits = ?, no_of_kits_picked = ?, status = ?,
code = ? WHERE id = ?") {

```
ps.setString(1, date);  
ps.setString(2, name);  
ps.setInt(3, no_of_kits);  
ps.setInt(4, no_of_kits_picked);  
ps.setString(5, status);  
ps.setInt(6, code);  
ps.setInt(7, id);
```

Here question mark places are replaced with the numbers like 1 means 1st question mark of query and at that place date must come.

This variables date,name,... Use in setInt is taken form volley request used in java file of Android studio.

The ones used un query is from database name of column.

Support all screen sizes

To support all screen sizes, no need to make any major changes only changes in XML file is required regarding constraint layout and it will be ready to use to any device, rest all code remains same for java and no play with id.

https://youtu.be/5lSQcJjZPFs?si=XL2woLu67txo_Zj-